

Hanging Print: Plastic Extrusion for Catenary Weaving in Mid-Air

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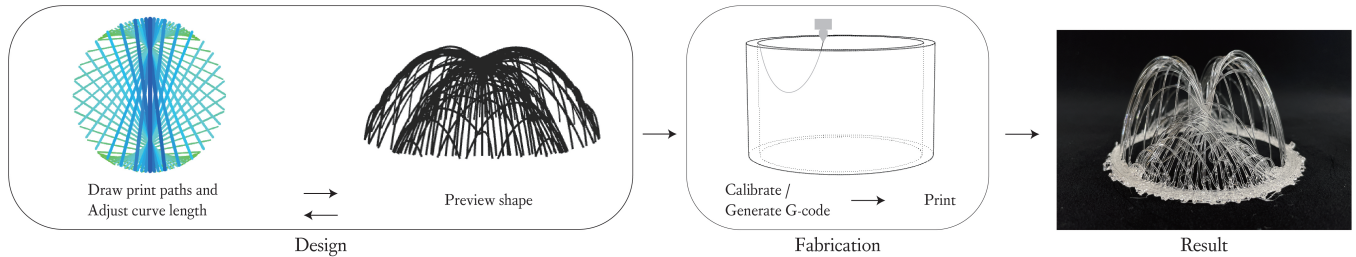


Figure 1: Workflow of Hanging Print

ABSTRACT

We present Hanging Print, a framework to design and fabricate shapes by weaving catenaries (i.e. hanging curves) by extruding plastic filaments directly in mid-air. Hanging Print has the potential to assist designing expressions unique to hanging structures, unique to plastic, and beyond imagination. In this paper we introduce the workflow of Hanging Print, and demonstrate our works in various scales.

CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.

KEYWORDS

3D printing, digital fabrication, computational design

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1 INTRODUCTION

As 3D printers such as Fused Filament Fabrication became more easily accessible, material extrusion technology has evolved into several printing methods that have been freed from the constraints of the traditional layered printing method. It is noteworthy that methods to print directly in mid-air have been proposed. Mueller [Mueller et al. 2014] proposed an approach print only the wireframe of the 3D model by preventing problems of deformation when printing a

line directly in mid-air, and made the approach applicable to others by implementing a software. On the other hand, some works such as [EmergingObjects 2016] intentionally exploit the phenomenon of deformation such as hanging as an aesthetic quality. These hanging shapes have been realized by empirical tests by actually printing them out, and the output shape has not been designed and predicted in advance.

In this paper, we present Hanging Print, a framework to design woven catenaries (i.e. hanging curves) by extruding plastic filaments directly in mid-air. We use our tool specialized in previewing the resulting shape and generating a g-code file (i.e. an instruction file for 3D printers to read) for Hanging Prints. Hanging Print has the potential to assist designing unique expressions of hanging. For example, spatial expressions with a large volume but small mass can be designed. Hanging Print also aims to bring out the potential of expression with plastic against strings. For example, the catenary shape could be maintained after the plastic hardens even when turned upside-down. Also, the woven catenary could be heated and fused together in post-processing to strengthen the structure. Lastly, Hanging Print aims to open the possibility of creating expressions beyond imagination. Exploring complexly overlapping curves woven into a curved surface may lead to finding a form one could not have imagined.

2 HANGING PRINT WORKFLOW

We implemented our tool with the Python interpreter component in Rhino Grasshopper. Firstly, our tool assists in speeding up the iteration between two processes: the adjustment of design and the preview the shape. Given a list of print paths and the desired length of each hanging curve, our tool approximates the resulting shape. The warping of curves when a hanging curve interferes with another is also simulated with our tool. Looking at the preview, users could change the position of the print paths and adjust the value of each curve's length.

Secondly, the information of path and length is converted to a g-code file. The details of conversion is explained in the following section. Some calibration is needed in this procedure. This is because

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the necessary volume of the catenary structure varies depending on the print environment since the die swell ratio and material's melt viscosity changes with the material and machine's property. The extrusion rate of the longest and shortest catenary structure should be tuned. The other curves' extrusion rate could be defined by interpolating these two values. Optionally, a vector data for cutting out pieces to assemble a platform that makes space to make hanging structures in mid-air can be generated.

Lastly, plastic extrusion would be performed on a platform lifted from the floor.

3 G-CODE CONVERSION

The length of the resulting catenary curve is mainly defined with the path's extrusion rate. The printing speed of each path is also adjusted to ensure the printing quality. If the extrusion rate changes while the print speed remains the same, the speed of the extruder motor will become irregular. As a result, the temperature of the filament becomes unstable and the accuracy decreases. Therefore, the more the desired catenary curve becomes longer, the higher the extrusion rate and the slower the print speed would be. Our conversion tool was developed according to this rule.

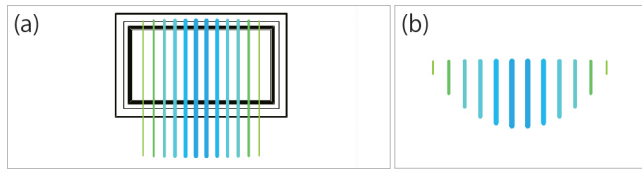


Figure 2: a)The path's print speed is expressed with color and the extrusion rate is expressed with thickness. As the speed gets faster, the color changes from blue to green. As the extrusion increases, the line gets thicker. b)Expected print result

4 RESULTS

4.1 Simple Catenary

The first work was made on a desktop size printer with a 0.8mm nozzle. The predicted shape was successfully reproduced. The structure maintained the shape and stood straight when flipped over. The second was printed on a 1M3 size machine with a 0.8mm nozzle. This work turned out to be very flexible and swayed in the wind. We used a multi-filament extruder, which gave a color-changing effect depending on the viewing angle. The third work was printed with a large-scale robot arm with a 6mm nozzle. This work did not succeed in printing the same shape as the preview. However, it should be noted that this structure also had the strength to stay standing when flipped over.

4.2 Catenary Weaving

We have also tested printing woven (i.e. overlapping) catenaries and made a complex curved surface. Our tool's simulation of the warping of interfering curves gave a fairly close preview of the printed result.

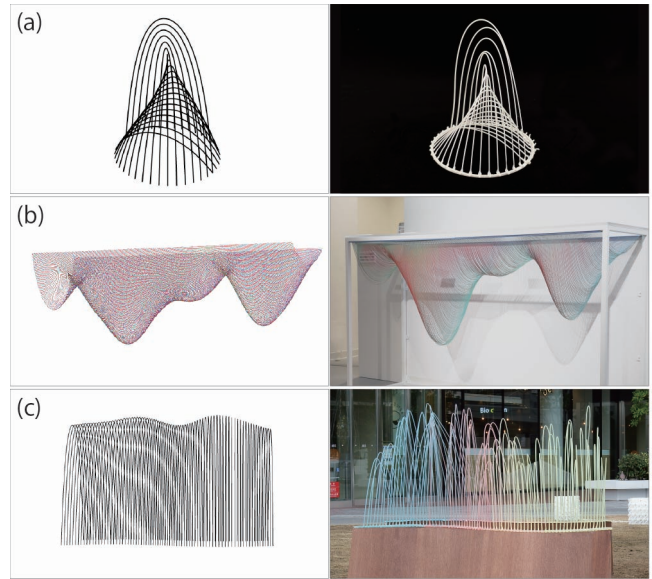


Figure 3: Previewed curves (left) and the printed results (right) in various sizes. The maximum depth of each work is a) 15cm, b)50cm, c)100cm



Figure 4: a)simple catenary b)simulated preview c)printed result

5 CONCLUSION AND DISCUSSION

In this paper we proposed Hanging Print, a method to design and fabricate hanging shapes with a 3D Printer. Through our study, we demonstrated that hanging shapes can be made in various scales. Moreover, hanging print has the potential to print beyond the machine's size. We also demonstrated that Hanging Prints could maintain their shape even when flipped upside-down. We demonstrated that woven catenaries create complex surfaces. We hope Hanging Print opens the possibility of creating expressions beyond imagination by exploiting natural phenomena as part of designing a shape.

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